

STA 2002

Lecture 3

Visualization (Briefly)

Estimation

Maximum likelihood estimator

Review of the last lecture

Moments of a distribution

Moment generating function

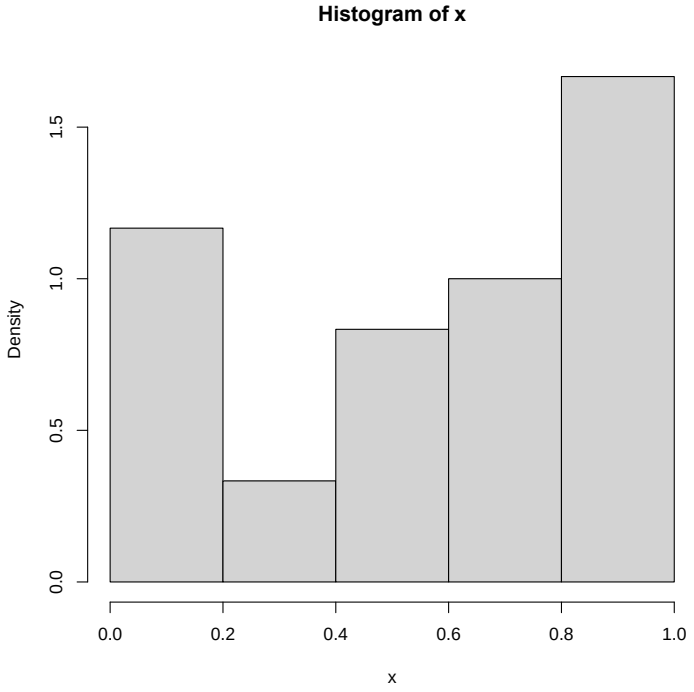
Central limit theorem

Bayes' rule

Time of midterm exam:
Oct 29 (Sunday) 9:30 am – 11:30 am

Data visualization: Histogram

```
0.97742091 0.99841003 0.48011321
0.08661486 0.58601176 0.61739409
0.65261468 0.38614097 0.87670884
0.73602044 0.16408397 0.02296603
0.43263809 0.64313709 0.31603898
0.90661059 0.67659943 0.70756372
0.92746488 0.81213553 0.95981676
0.84393371 0.45332443 0.13658223
0.03843230 0.83956954 0.03093576
0.50768522 0.81352515 0.11346368
```



We shall denote these k class intervals by

$$(c_0, c_1], (c_1, c_2], \ldots, (c_{k-1}, c_k].$$

For each interval, find the relative frequency or density of data points falling into the interval

$$h(x) = \frac{f_i}{(n)(c_i - c_{i-1})}, \qquad \text{for } c_{i-1} < x \leq c_i, \quad i = 1, 2, \ldots, k,$$

Area equal to the relative frequency $\frac{f_i}{n}$

Data visualization: Histogram

Different bin sizes reveal different detail levels

Too small: artificial peaks and valleys

Too large: Too little detail

The proper number of bins depends on how many data points you have, as well as the characteristics of the distribution.

Usually programs can choose automatically based on some rules.

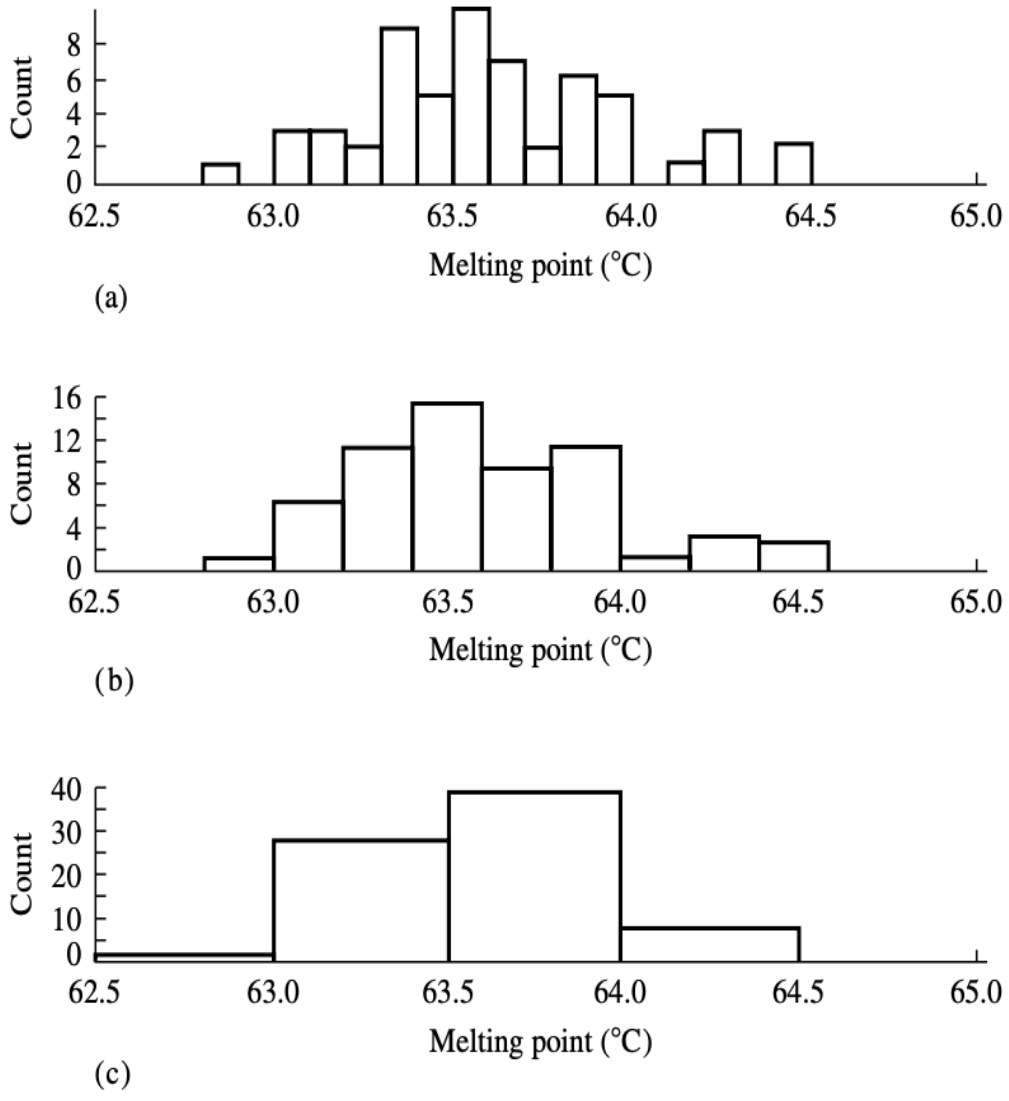


FIGURE **10.8** Histograms of melting points of beeswax: (a) bin width = .1, (b) bin width = .2, (c) bin width = .5.

Data visualization: Histogram

Can use uneven bins when the data is skewed and some regions contain too few data points.

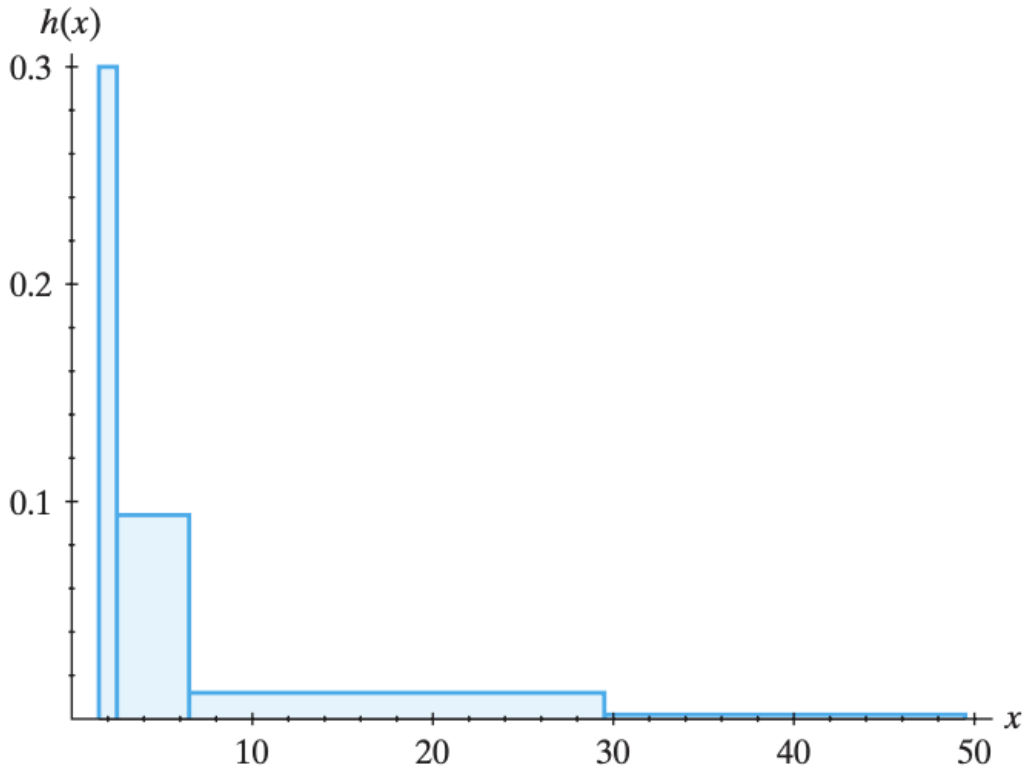


Figure 6.1-3 Relative frequency histogram of losses

Data visualization: Stem-and-leaf plots

63.78	63.45	63.58	63.08	63.40	64.42	63.27	63.10
63.34	63.50	63.83	63.63	63.27	63.30	63.83	63.50
63.36	63.86	63.34	63.92	63.88	63.36	63.36	63.51
63.51	63.84	64.27	63.50	63.56	63.39	63.78	63.92
63.92	63.56	63.43	64.21	64.24	64.12	63.92	63.53
63.50	63.30	63.86	63.93	63.43	64.40	63.61	63.03
63.68	63.13	63.41	63.60	63.13	63.69	63.05	62.85
63.31	63.66	63.60					



	STEM	LEAF
1	628	:5
0	629	:
3	630	:358
3	631	:033
2	632	:77
9	633	:001446669
5	634	:01335
10	635	:0000113668
7	636	:0013689
2	637	:88
6	638	:334668
5	639	:22223
0	640	:
1	641	:2
3	642	:147
0	643	:
2	644	:02

Data visualization: Boxplot

Range = difference of the extremes.

Interquartile range = difference of third and first quartiles

Midrange = average of the extremes

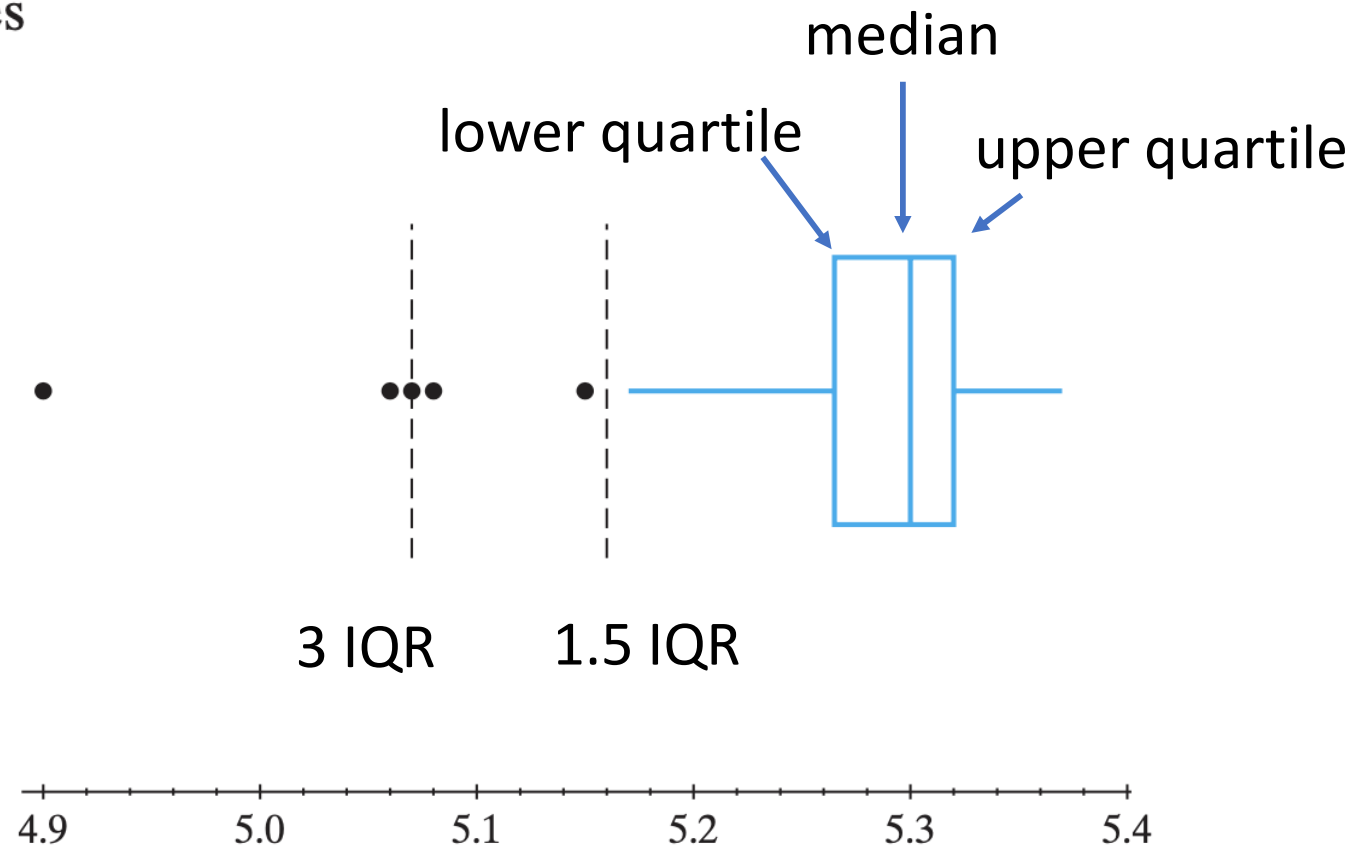


Figure 6.2-3 Box plot for weights of 39 gold coins with fences and outliers

Data visualization: Boxplot

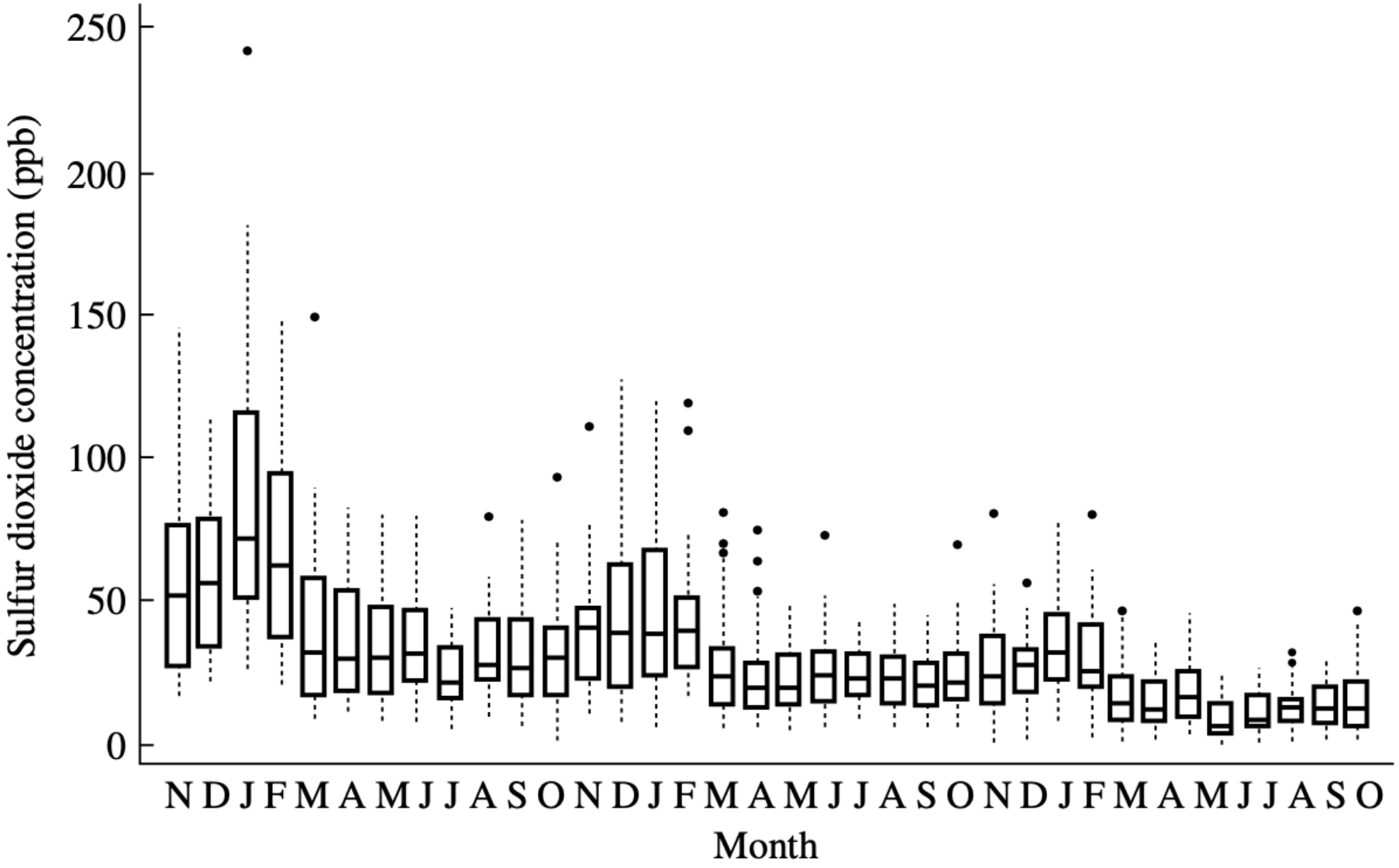


FIGURE **10.14** Boxplots of daily maximum concentrations of sulfur dioxide.

The general setup

Q: We have a data set, and we want to **infer** the properties of the underlying distribution from this data set.

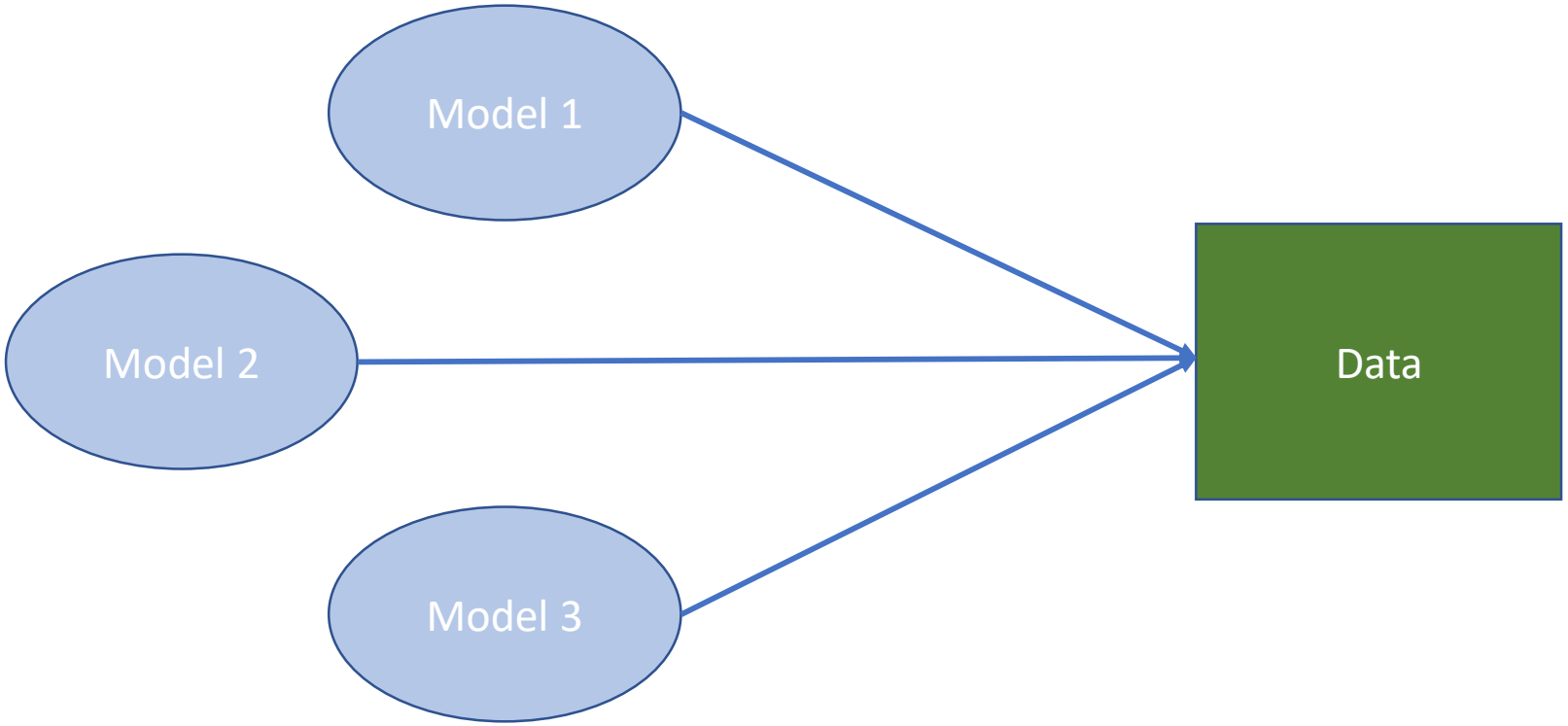
A: **Inductive reasoning** rather than **deductive reasoning -** a variety of different probability models must at least be explored to see which model best “fits” the data.



The general setup

Q: We have a data set, and we want to **infer** the properties of the underlying distribution from this data set.

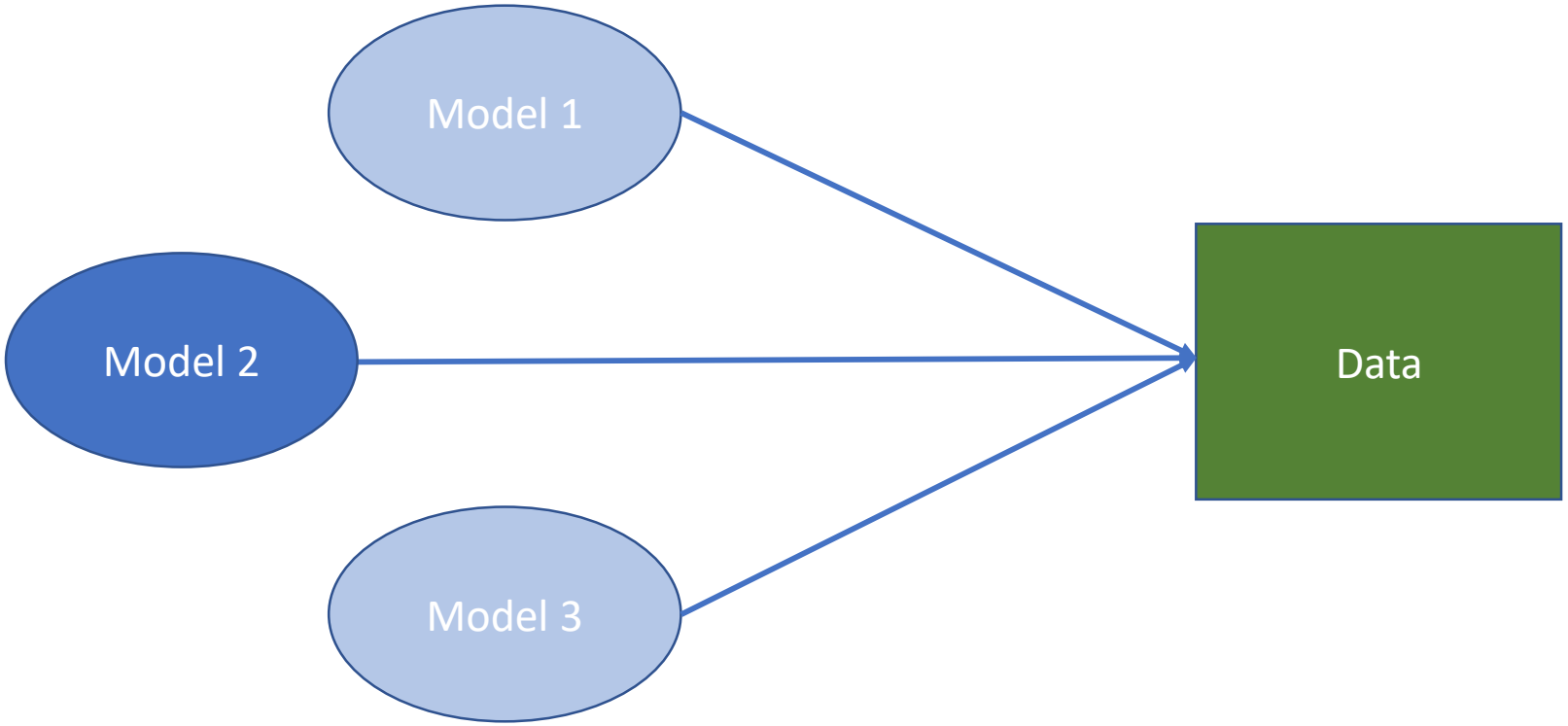
A: I**nductive reasoning** rather than **deductive reasoning -** a variety of different probability models must at least be explored to see which model best “fits” the data.



The general setup

Q: We have a data set, and we want to **infer** the properties of the underlying distribution from this data set.

A: I**nductive reasoning** rather than **deductive reasoning -** a variety of different probability models must at least be explored to see which model best “fits” the data.



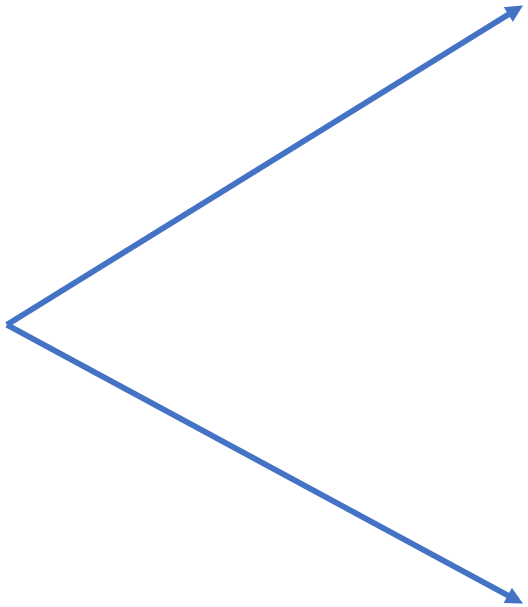
The general setup

BASIS FOR COMPARISON	INDUCTIVE REASONING	DEDUCTIVE REASONING
Meaning	Inductive Reasoning connotes the argument in which the premises give reasons in support of the probable truth of the conjecture.	Deductive reasoning is the fundamental form of valid reasoning, wherein the premises give guarantee of the truth of conjecture.
Approach	Bottom-up approach	Top-down approach
Starting point	Conclusion	Premises
Based on	Patterns or trend	Facts, truths and rules
Process	Observation > Pattern > Tentative Hypothesis > Theory	Theory > Hypothesis > Observation > Confirmation
Argument	May or may not be strong.	May or may not be valid.

https://keydifferences.com/difference-between-inductive-and-deductive-reasoning.html

The general setup

Statistical inference



Estimation

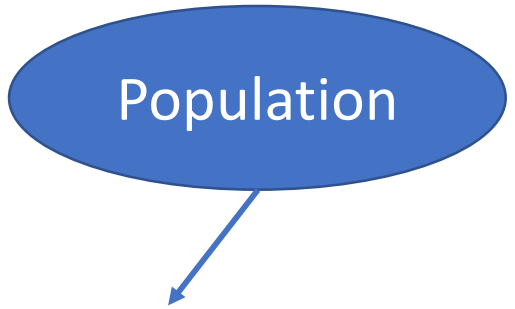
Estimating the values of specific population parameters .

Hypothesis testing

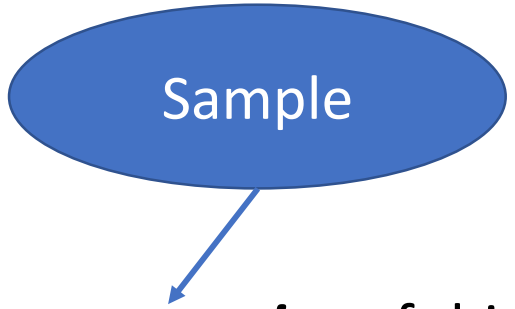
Testing whether the value of a population parameter is equal to some specific value.

The general setup

Population and Sample



Goal: characterize the distribution of birthweights of **all liveborn infants** born in the United States in 2008



Method: select a **random sample of *n* infants who are *representative*** of this large group and use the birthweights $x_1, \ldots, x_n$ from this sample to help us estimate $\mu \; and \; \sigma^2$

Sometimes the by "population" we mean an underlying distribution.

The general setup

A random sample is a selection of some members of the population such that each member is independently chosen and has a known nonzero probability of being selected.

A simple random sample is a random sample in which each group member has the same probability of being selected.

The general setup

A random sample is a selection of some members of the population such that each member is independently chosen and has a known nonzero probability of being selected.

A simple random sample is a random sample in which each group member has the same probability of being selected.

Ex: A clinical trial for the efficacy of a drug

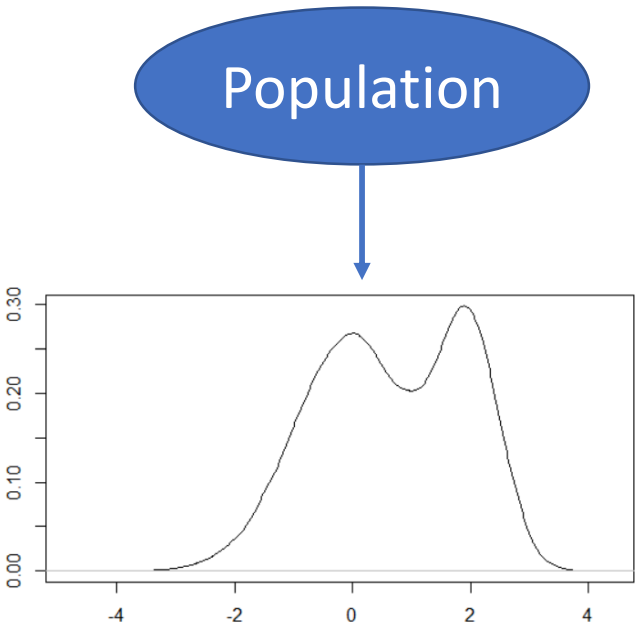
Population: all people who have the disease

A simple random sample is impossible – need to consider every small clinic, or even diseased people who don't go to hospital

A (practical) "random" sample: people who go to some large medical centers that agree to participate in the clinical trial

Assumption: these people are **representative** of the population

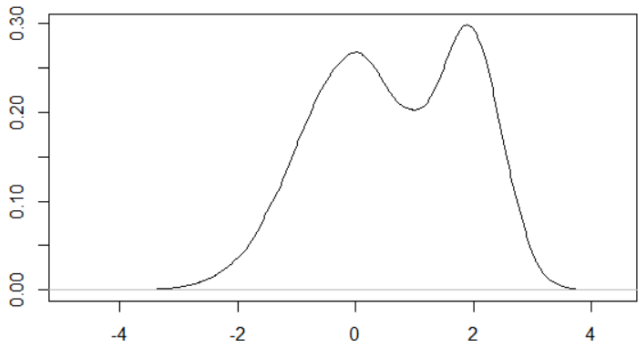
The “frequentist” thinking



$$\mu = 0.66$$
$$\sigma^2 = 1.65$$

The “frequentist” thinking

Population



$$\mu = 0.66$$
$$\sigma^2 = 1.65$$

Samples

-1.28	1.81	1.45	0.05	2.08	-1.15	-0.34	2.23	-0.20	1.20
1.04	-1.12	1.88	-0.86	0.25	2.39	-0.76	0.64	2.54	0.54
1.67	1.91	1.72	2.17	0.15	-1.24	2.42	-0.53	-1.36	-0.56

-0.74	1.62	-0.14	-0.67	2.01	1.66	1.42	-0.70	2.61	0.46
-0.68	-0.85	1.51	1.96	0.79	-0.37	-1.44	-0.33	1.50	-1.13
1.48	1.27	0.28	0.54	0.63	-0.44	-0.02	0.41	1.52	1.48

-0.42	0.65	0.25	-2.76	2.78	-1.10	1.36	1.11	0.56	2.29
2.89	0.75	0.02	1.58	-1.02	1.35	-0.57	1.32	-0.46	1.62
-0.13	-0.36	-2.33	1.73	1.23	2.07	0.92	2.01	-0.36	-0.36

-0.20	1.32	2.46	1.84	-0.31	0.77	-0.13	-0.04	-1.93	2.09
2.86	2.34	-0.41	0.41	2.86	1.50	1.80	0.68	0.12	0.61
1.39	-0.14	-0.36	0.35	-0.08	0.96	1.95	-2.16	2.04	1.44

1.13	0.08	1.76	-0.35	0.09	2.67	2.76	0.57	-0.40	1.66
0.01	-0.16	-1.24	2.32	-0.66	-0.03	-1.44	-0.88	0.59	2.02
2.90	-0.05	-0.83	-0.45	0.33	-1.02	-0.33	1.40	1.93	2.04

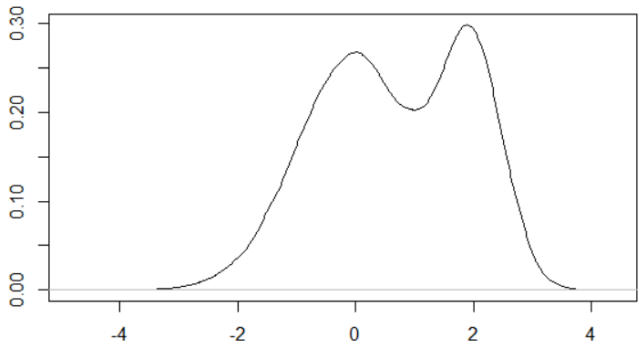
1.60	-0.32	1.66	2.00	2.24	-0.05	1.48	1.18	-0.42	2.07
-0.36	2.33	0.33	-0.29	-0.18	-1.36	1.66	3.00	-1.04	-1.53
0.90	-0.36	-0.80	2.43	-0.93	2.17	0.14	1.84	-1.16	-0.33

0.09	0.19	-0.95	1.95	1.41	0.78	1.75	-0.40	-0.83	-0.50
1.19	0.51	-1.10	-1.17	2.09	-0.57	1.56	0.39	0.95	-0.69
2.74	-0.08	2.38	2.36	2.50	-1.02	1.94	2.20	2.10	-0.25

1.98	-0.07	2.40	-0.16	0.47	0.44	-0.82	-0.49	-0.23	0.07
1.86	1.43	-0.15	1.70	2.40	0.49	-0.71	0.60	-0.58	0.76
0.47	1.95	0.64	1.76	-0.02	2.16	-0.72	2.81	1.48	0.04

The “frequentist” thinking

Population



$$\mu = 0.66$$
$$\sigma^2 = 1.65$$

Samples

Sample mean	Sample variance
0.62	1.78
0.52	1.20
0.55	1.91
0.80	1.62
0.54	1.67
0.59	1.80
0.71	1.65
0.73	1.17

```
-1.28  1.81  1.45  0.05  2.08 -1.15 -0.34  2.23 -0.20  1.20
 1.04 -1.12  1.88 -0.86  0.25  2.39 -0.76  0.64  2.54  0.54
 1.67  1.91  1.72  2.17  0.15 -1.24  2.42 -0.53 -1.36 -0.56
```

```
-0.74  1.62 -0.14 -0.67  2.01  1.66  1.42 -0.70  2.61  0.46
-0.68 -0.85  1.51  1.96  0.79 -0.37 -1.44 -0.33  1.50 -1.13
 1.48  1.27  0.28  0.54  0.63 -0.44 -0.02  0.41  1.52  1.48
```

```
-0.42  0.65  0.25 -2.76  2.78 -1.10  1.36  1.11  0.56  2.29
 2.89  0.75  0.02  1.58 -1.02  1.35 -0.57  1.32 -0.46  1.62
-0.13 -0.36 -2.33  1.73  1.23  2.07  0.92  2.01 -0.36 -0.36
```

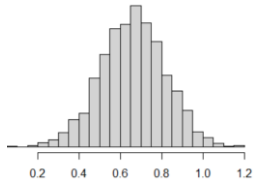
```
-0.20  1.32  2.46  1.84 -0.31  0.77 -0.13 -0.04 -1.93  2.09
 2.86  2.34 -0.41  0.41  2.86  1.50  1.80  0.68  0.12  0.61
 1.39 -0.14 -0.36  0.35 -0.08  0.96  1.95 -2.16  2.04  1.44
```

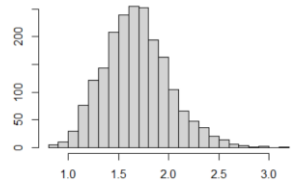
```
 1.13  0.08  1.76 -0.35  0.09  2.67  2.76  0.57 -0.40  1.66
 0.01 -0.16 -1.24  2.32 -0.66 -0.03 -1.44 -0.88  0.59  2.02
 2.90 -0.05 -0.83 -0.45  0.33 -1.02 -0.33  1.40  1.93  2.04
```

```
 1.60 -0.32  1.66  2.00  2.24 -0.05  1.48  1.18 -0.42  2.07
-0.36  2.33  0.33 -0.29 -0.18 -1.36  1.66  3.00 -1.04 -1.53
 0.90 -0.36 -0.80  2.43 -0.93  2.17  0.14  1.84 -1.16 -0.33
```

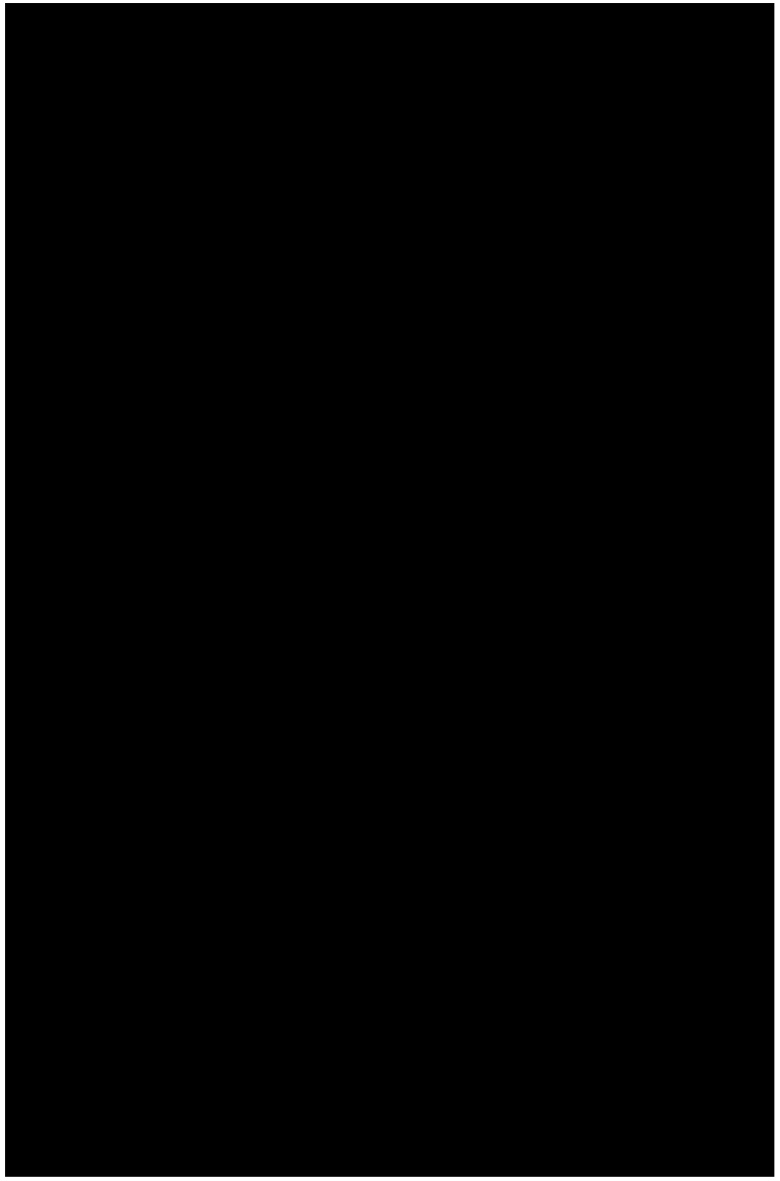
```
 0.09  0.19 -0.95  1.95  1.41  0.78  1.75 -0.40 -0.83 -0.50
 1.19  0.51 -1.10 -1.17  2.09 -0.57  1.56  0.39  0.95 -0.69
 2.74 -0.08  2.38  2.36  2.50 -1.02  1.94  2.20  2.10 -0.25
```

```
 1.98 -0.07  2.40 -0.16  0.47  0.44 -0.82 -0.49 -0.23  0.07
 1.86  1.43 -0.15  1.70  2.40  0.49 -0.71  0.60 -0.58  0.76
 0.47  1.95  0.64  1.76 -0.02  2.16 -0.72  2.81  1.48  0.04
```





The “frequentist” thinking



Samples

-1.28	1.81	1.45	0.05	2.08	-1.15	-0.34	2.23	-0.20	1.20
1.04	-1.12	1.88	-0.86	0.25	2.39	-0.76	0.64	2.54	0.54
1.67	1.91	1.72	2.17	0.15	-1.24	2.42	-0.53	-1.36	-0.56

-0.74	1.62	-0.14	-0.67	2.01	1.66	1.42	-0.70	2.61	0.46
-0.68	-0.85	1.51	1.96	0.79	-0.37	-1.44	-0.33	1.50	-1.13
1.48	1.27	0.28	0.54	0.63	-0.44	-0.02	0.41	1.52	1.48

-0.42	0.65	0.25	-2.76	2.78	-1.10	1.36	1.11	0.56	2.29
2.89	0.75	0.02	1.58	-1.02	1.35	-0.57	1.32	-0.46	1.62
-0.13	-0.36	-2.33	1.73	1.23	2.07	0.92	2.01	-0.36	-0.36

-0.20	1.32	2.46	1.84	-0.31	0.77	-0.13	-0.04	-1.93	2.09
2.86	2.34	-0.41	0.41	2.86	1.50	1.80	0.68	0.12	0.61
1.39	-0.14	-0.36	0.35	-0.08	0.96	1.95	-2.16	2.04	1.44

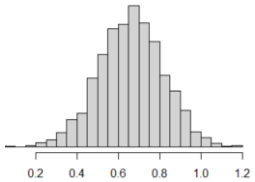
1.13	0.08	1.76	-0.35	0.09	2.67	2.76	0.57	-0.40	1.66
0.01	-0.16	-1.24	2.32	-0.66	-0.03	-1.44	-0.88	0.59	2.02
2.90	-0.05	-0.83	-0.45	0.33	-1.02	-0.33	1.40	1.93	2.04

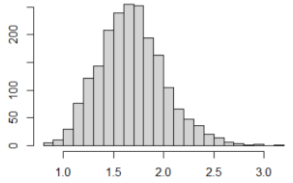
1.60	-0.32	1.66	2.00	2.24	-0.05	1.48	1.18	-0.42	2.07
-0.36	2.33	0.33	-0.29	-0.18	-1.36	1.66	3.00	-1.04	-1.53
0.90	-0.36	-0.80	2.43	-0.93	2.17	0.14	1.84	-1.16	-0.33

0.09	0.19	-0.95	1.95	1.41	0.78	1.75	-0.40	-0.83	-0.50
1.19	0.51	-1.10	-1.17	2.09	-0.57	1.56	0.39	0.95	-0.69
2.74	-0.08	2.38	2.36	2.50	-1.02	1.94	2.20	2.10	-0.25

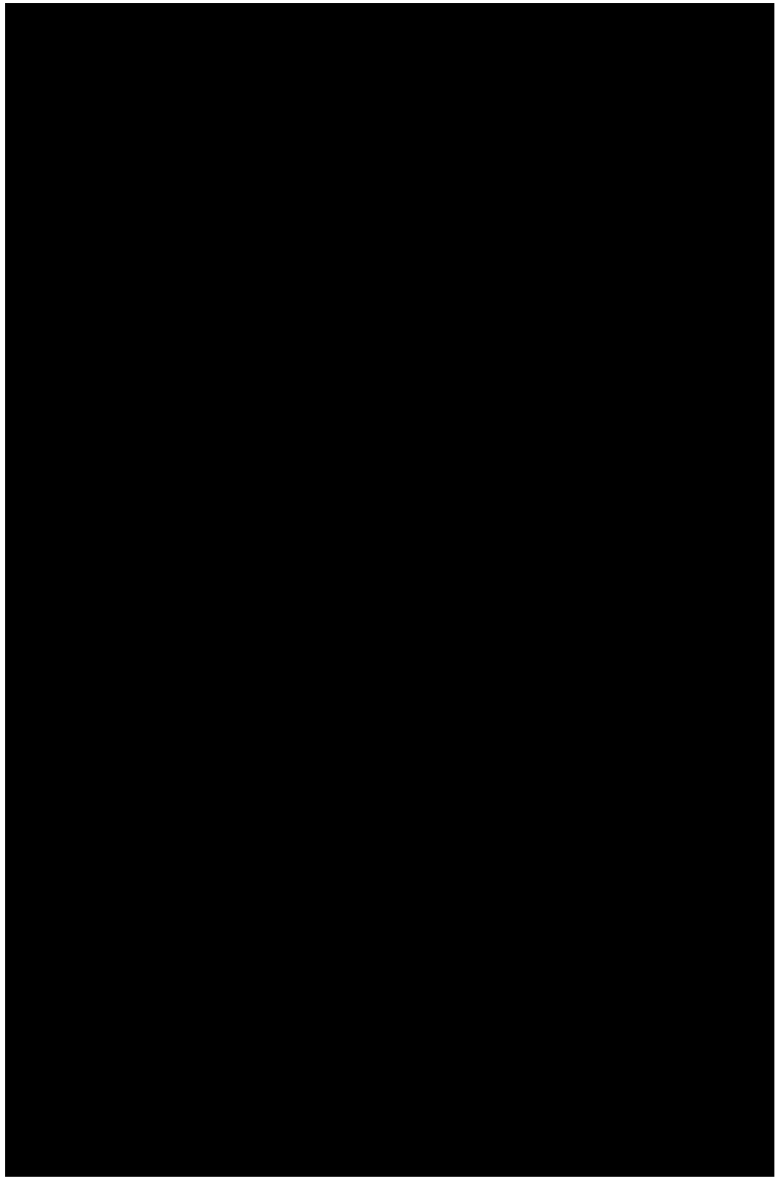
1.98	-0.07	2.40	-0.16	0.47	0.44	-0.82	-0.49	-0.23	0.07
1.86	1.43	-0.15	1.70	2.40	0.49	-0.71	0.60	-0.58	0.76
0.47	1.95	0.64	1.76	-0.02	2.16	-0.72	2.81	1.48	0.04

Sample mean	Sample variance
0.62	1.78
0.52	1.20
0.55	1.91
0.80	1.62
0.54	1.67
0.59	1.80
0.71	1.65
0.73	1.17



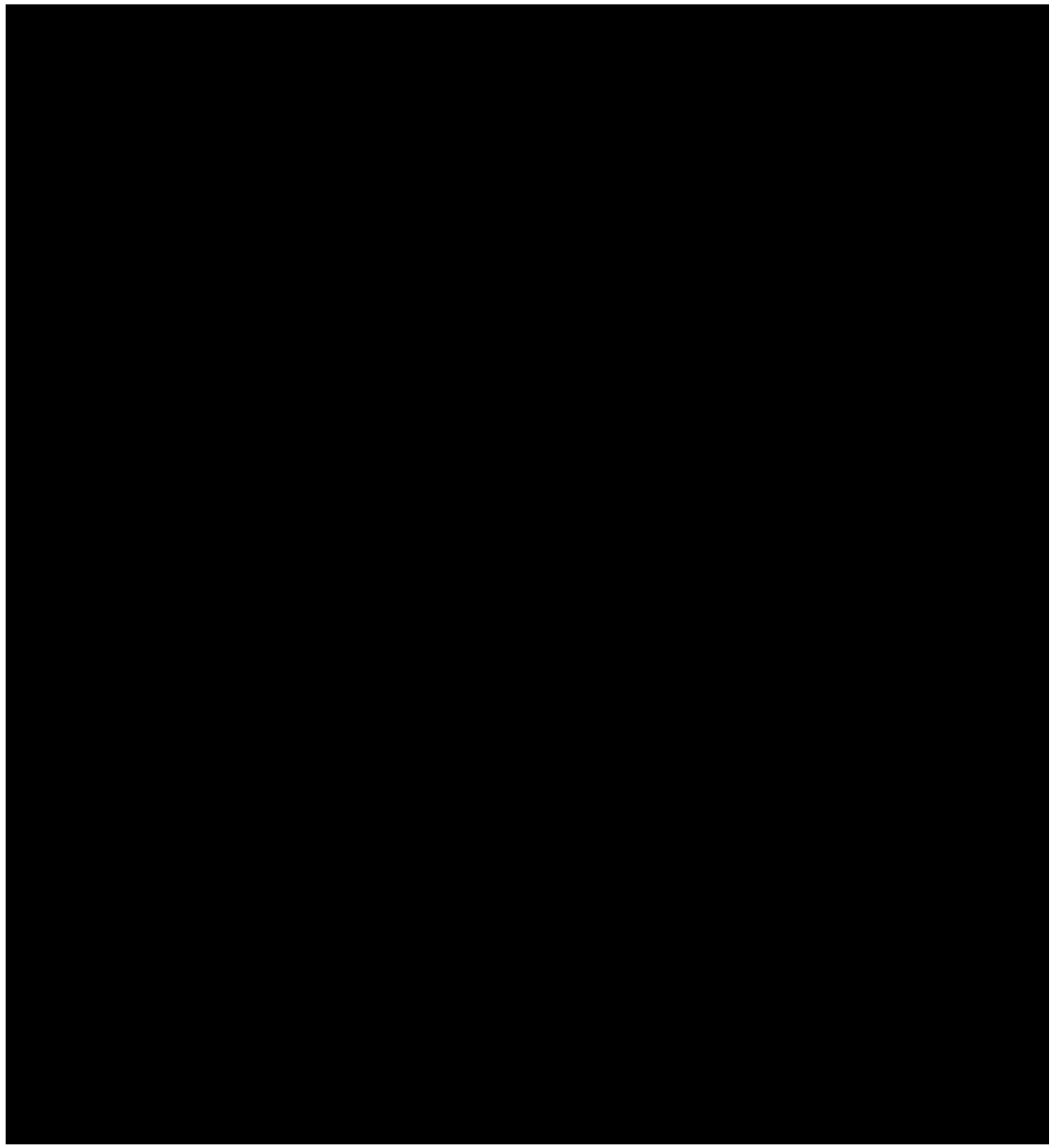


The “frequentist” thinking

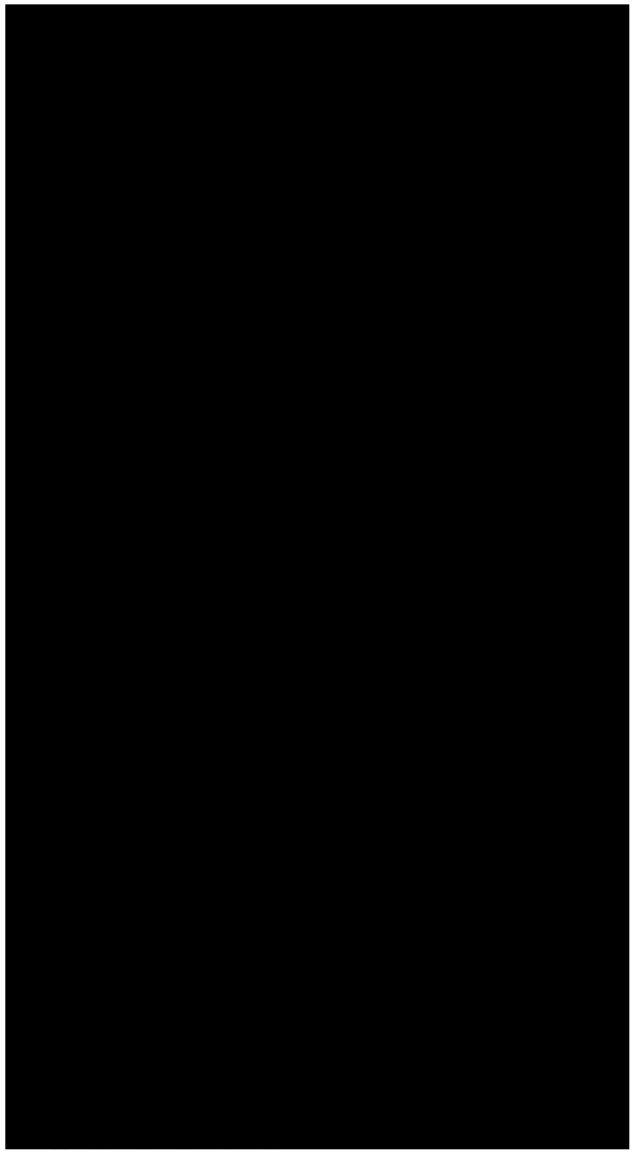


Samples

-1.28	1.81	1.45	0.05	2.08	-1.15	-0.34	2.23	-0.20	1.20
1.04	-1.12	1.88	-0.86	0.25	2.39	-0.76	0.64	2.54	0.54
1.67	1.91	1.72	2.17	0.15	-1.24	2.42	-0.53	-1.36	-0.56



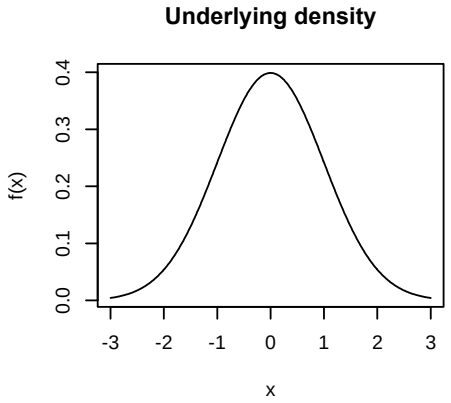
Sample mean	Sample variance
0.62	1.78

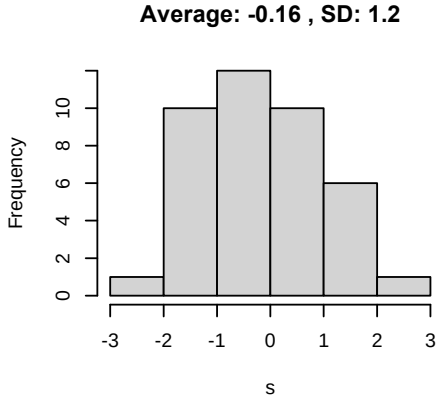


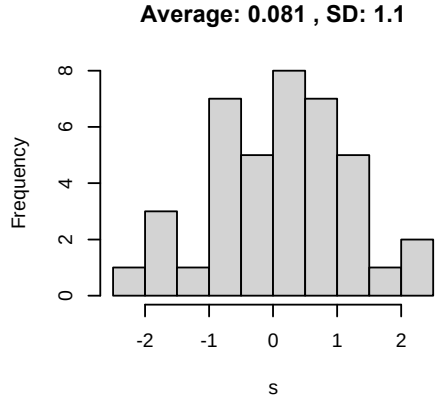
The general setup

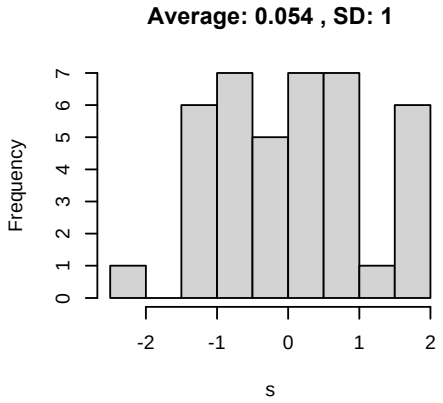
What “uncertainty” are we trying to address in the inference?

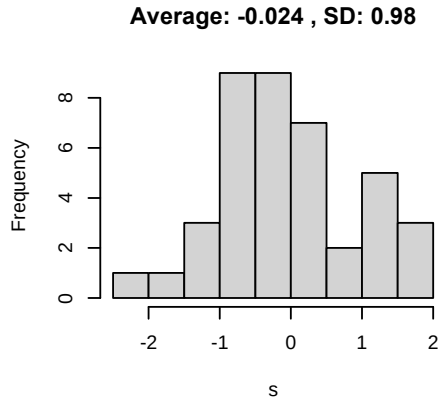
Sampling variation.

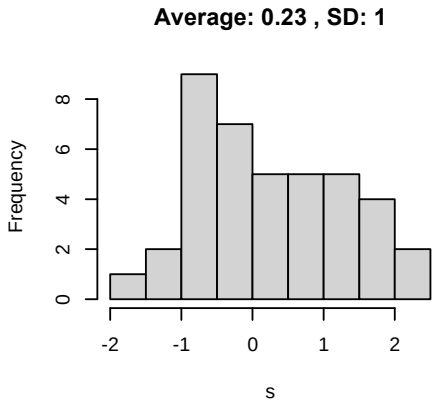


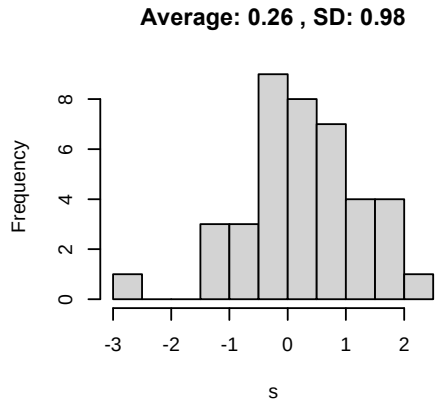












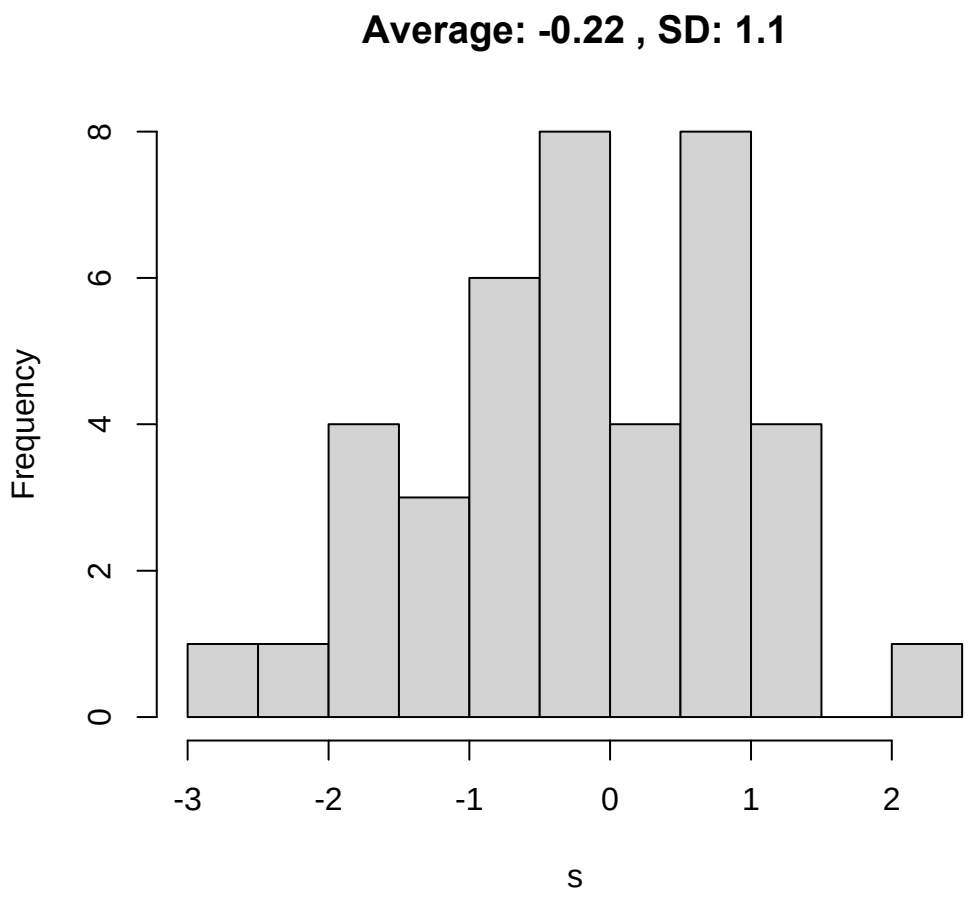
The general setup

In reality, we have just a single random sample with size n.

Is this from a normal distribution? Or can it be from a t-distribution? Or a uniform distribution?

If it is from a normal distribution, how uncertain are we about our estimation of the population parameters $\mu \; and \; \sigma^2$?

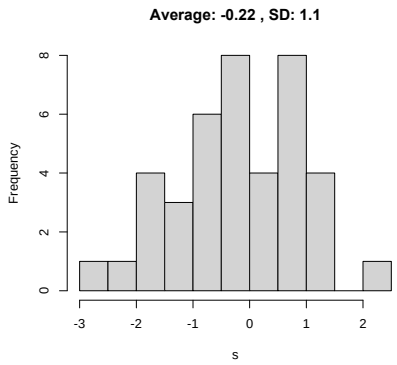
We have to be able to assign uncertainties in parameter estimation, and compare which model is more likely.



An over-simplified example to get the taste

Data

```
-0.410284714 -0.763757008  1.266399194
 1.384219688  0.718491040  0.776665109
 0.302378038 -0.370678878 -0.698984387
-1.816156611  1.168388569 -0.519340753
-2.011902398 -0.420420349  0.006307144
 0.602828400  0.628296684 -2.529556727
-0.696027668  0.592583349 -1.362589998
 2.012336774 -0.075041237 -1.387026404
 0.525074294 -1.612665456 -0.154134509
-0.154612913 -1.763669499 -0.595799838
 0.788510864 -1.066526201 -0.370630741
 1.214197222  0.194723107 -1.837446679
 0.561737185 -0.943725024 -0.124440443
 0.071928824
```



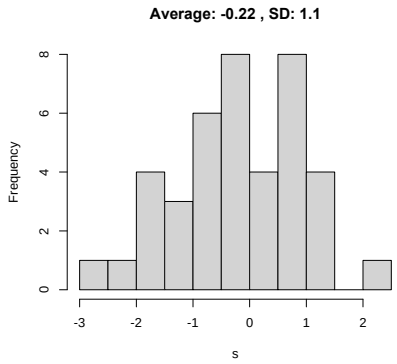
Normal:
most likely parameters
$\hat{\mu} = -0.22, \hat{\sigma}^2 = 1.12$ → Likelihood of generating the sample from the distribution: $10^{-25.4}$

Uniform:
most likely parameters
min=-2.5, max=2.0 → Likelihood of generating the sample from the distribution: 10^{-31}

An over-simplified example to get the taste

Data

```
-0.410284714 -0.763757008  1.266399194
 1.384219688  0.718491040  0.776665109
 0.302378038 -0.370678878 -0.698984387
-1.816156611  1.168388569 -0.519340753
-2.011902398 -0.420420349  0.006307144
 0.602828400  0.628296684 -2.529556727
-0.696027668  0.592583349 -1.362589998
 2.012336774 -0.075041237 -1.387026404
 0.525074294 -1.612665456 -0.154134509
-0.154612913 -1.763669499 -0.595799838
 0.788510864 -1.066526201 -0.370630741
 1.214197222  0.194723107 -1.837446679
 0.561737185 -0.943725024 -0.124440443
 0.071928824
```



Normal:
most likely parameters
$\hat{\mu} = -0.22, \hat{\sigma}^2 = 1.12$ → Likelihood of generating the sample from the distribution: $10^{-25.4}$

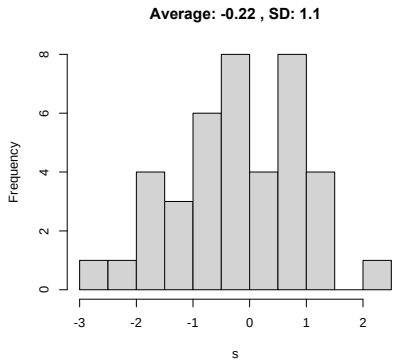
Uniform:
most likely parameters
min=-2.5, max=2.0 → Likelihood of generating the sample from the distribution: 10^{-31}

An over-simplified example to get the taste

Likelihood of generating the sample from the distribution:

Data

```
-0.410284714 -0.763757008  1.266399194
 1.384219688  0.718491040  0.776665109
 0.302378038 -0.370678878 -0.698984387
-1.816156611  1.168388569 -0.519340753
-2.011902398 -0.420420349  0.006307144
 0.602828400  0.628296684 -2.529556727
-0.696027668  0.592583349 -1.362589998
 2.012336774 -0.075041237 -1.387026404
 0.525074294 -1.612665456 -0.154134509
-0.154612913 -1.763669499 -0.595799838
 0.788510864 -1.066526201 -0.370630741
 1.214197222  0.194723107 -1.837446679
 0.561737185 -0.943725024 -0.124440443
 0.071928824
```



Normal: $\mu = -0.22, \sigma^2 = 1.12$ $10^{-25.4}$

Normal: $\mu = 0, \sigma^2 = 1$ $10^{-25.8}$

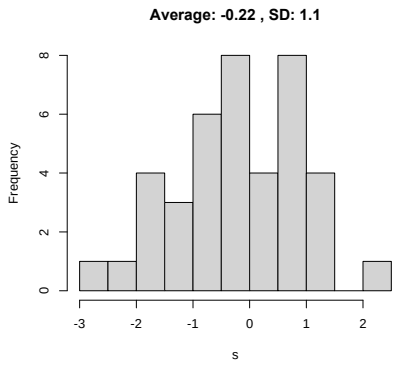
Normal: $\mu = -1, \sigma^2 = 2$ $10^{-31.7}$

An over-simplified example to get the taste

Likelihood of generating the sample from the distribution:

Data

```
-0.410284714 -0.763757008  1.266399194
 1.384219688  0.718491040  0.776665109
 0.302378038 -0.370678878 -0.698984387
-1.816156611  1.168388569 -0.519340753
-2.011902398 -0.420420349  0.006307144
 0.602828400  0.628296684 -2.529556727
-0.696027668  0.592583349 -1.362589998
 2.012336774 -0.075041237 -1.387026404
 0.525074294 -1.612665456 -0.154134509
-0.154612913 -1.763669499 -0.595799838
 0.788510864 -1.066526201 -0.370630741
 1.214197222  0.194723107 -1.837446679
 0.561737185 -0.943725024 -0.124440443
 0.071928824
```



Normal: $\mu = -0.22, \sigma^2 = 1.12$ $10^{-25.4}$

Normal: $\mu = 0, \sigma^2 = 1$ $10^{-25.8}$

Normal: $\mu = -1, \sigma^2 = 2$ $10^{-31.7}$

We can't exhaust all possibilities.

How do we find this best parameter?

It is an **optimization problem**

Maximum Likelihood

Given a model, we want to find the most likely parameter setting given we observe the data, i.e. the θ that maximizes $P(\theta \mid Data)$

This is equivalent to finding the parameter setting that give the highest likelihood of the data: $P(Data \mid \theta)$, if we don't hold any prior belief about θ.

Maximum Likelihood

Given a model, we want to find the most likely parameter setting given we observe the data, i.e. the θ that maximizes $P(\theta \mid Data)$

This is equivalent to finding the parameter setting that give the highest likelihood of the data: $P(Data \mid \theta)$, if we don't hold any prior belief about θ.

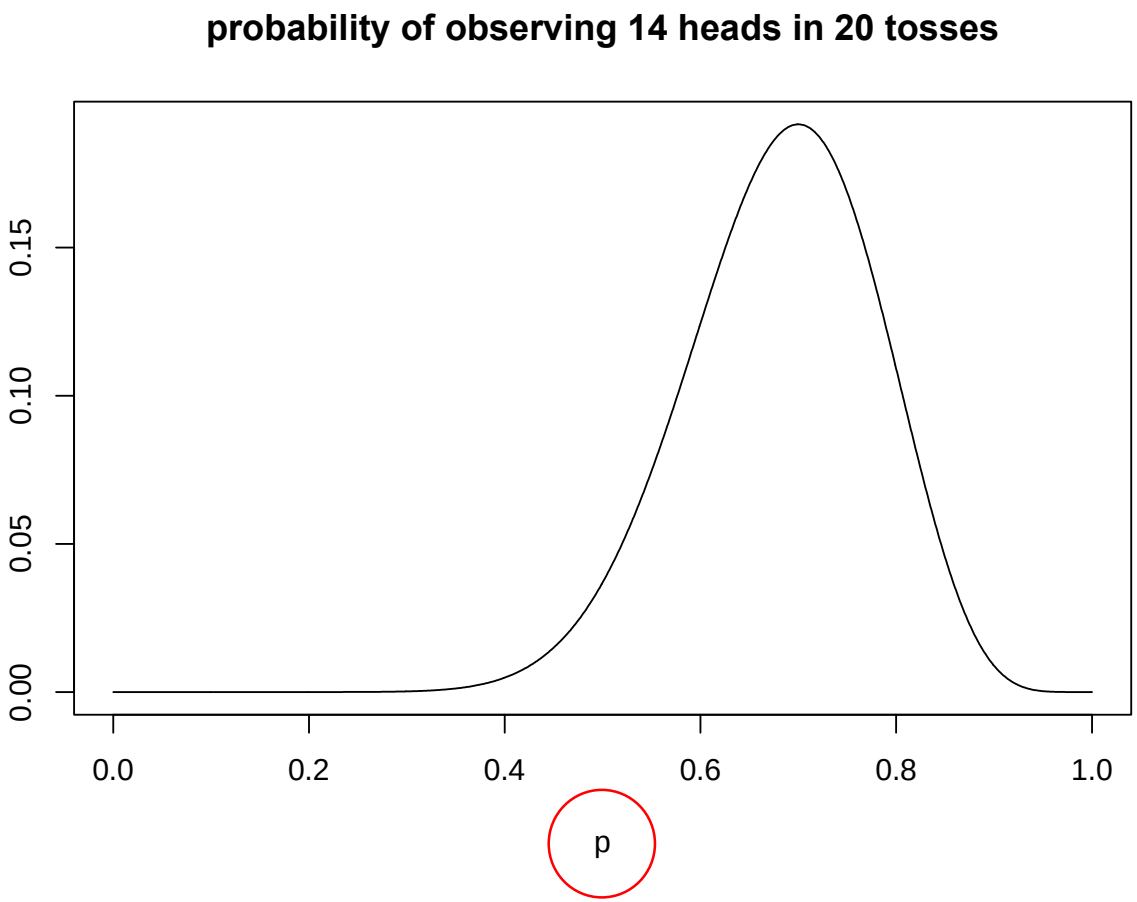
$$P(\theta \mid Data) = \frac{P(Data \mid \theta)P(\theta)}{\sum_{\lambda \in \Theta} P(Data \mid \lambda)P(\lambda)}$$

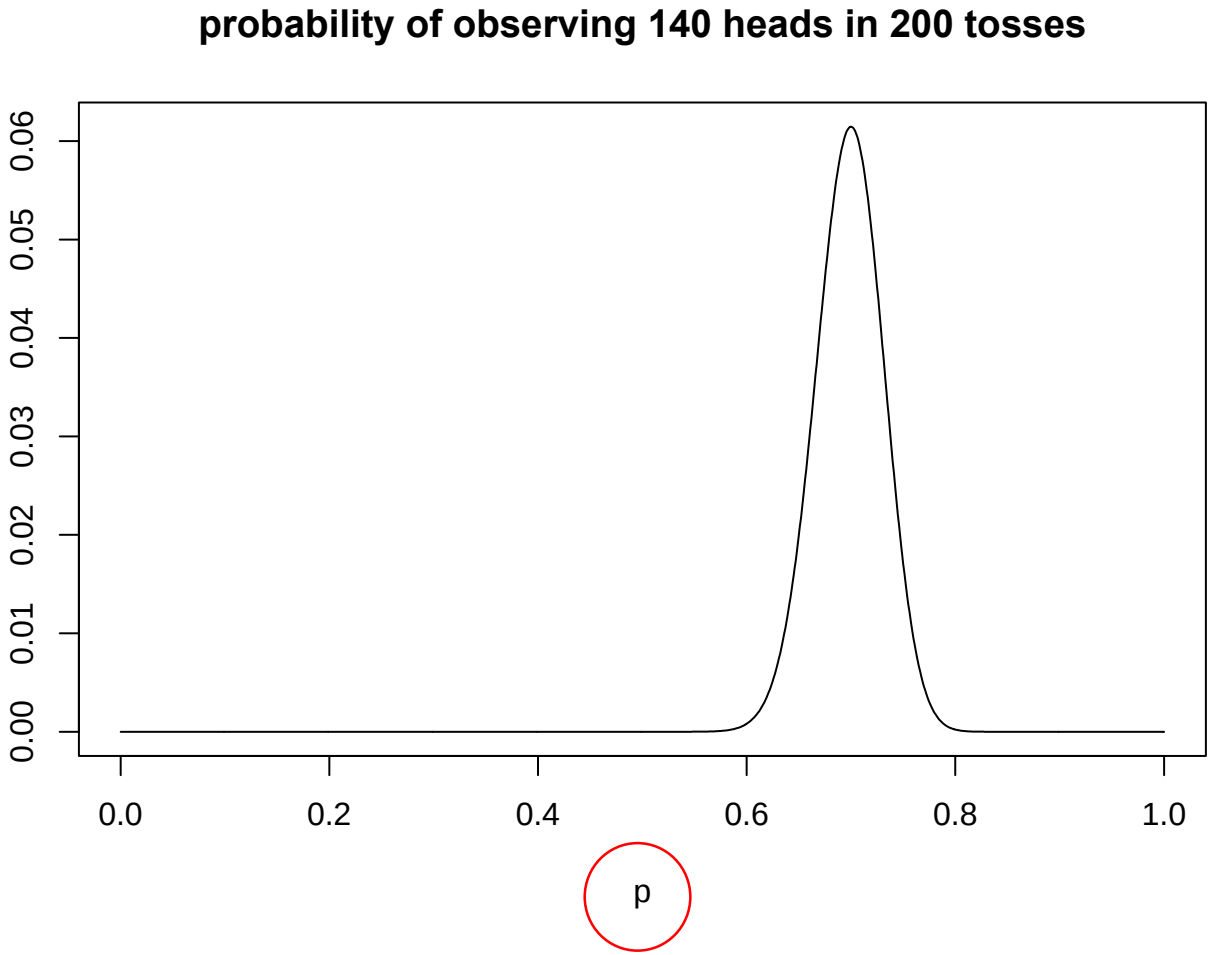
Stays the same for every possible θ

Summing over all possible $\boldsymbol{\theta}$, stays the same.

Maximum Likelihood

A binomial example:





Maximum Likelihood

Note: probability isn't equal to likelihood (using density function) in the continuous case

Ex: $X \sim N(0,1)$
$f(0)=0.4$

This isn't saying there is a 40% probability of observing "0" when sampling from N(0,1). In fact, we almost never observe exact "0". But we can observe a lot of points around 0.

In relative terms, still $f(0) >> f(2)$ when $X \sim N(0,1)$. This means we are much more likely to get a data point in the small vicinity of 0, as compared to in the small vicinity of 2.

So in spirit the previous argument still holds.

Maximum Likelihood

Random variables $X_1, \ldots, X_n$ have a joint density or frequency function $f(x_1,x_2,\ldots,x_n|\theta)$.

Observed values $X_i = x_i$, where $i = 1,\ldots,n$

The likelihood of θ as a function of $x_1,x_2,\ldots,x_n$ is defined as

$$L(\theta) = f(x_1, x_2, \cdots, x_n|\theta)$$

The **maximum likelihood estimate (MLE)** of θ is that value of θ that maximizes the likelihood— makes the observed data "most probable" or "most likely."

Maximum Likelihood

Random variables $X_1, \ldots, X_n$ have a joint density or frequency function $f(x_1,x_2,\ldots,x_n|\theta)$.

Observed values $X_i = x_i$, where $i = 1,\ldots,n$

The likelihood of θ as a function of $x_1,x_2,\ldots,x_n$ is defined as

$$L(\theta) = f(x_1, x_2, \cdots, x_n|\theta)$$

The **maximum likelihood estimate (MLE)** of θ is that value of θ that maximizes the likelihood— makes the observed data "most probable" or "most likely."

If the X_i are assumed to be i.i.d., $$L(\theta) = \prod_{i=1}^{n} f(X_i|\theta)$$

Maximum Likelihood

It is usually easier to maximize the natural logarithm of the likelihood

Easier analytical form,

Better numerical stability – the likelihood easily underflows when n is large.

Maximizing log-likelihood is equivalent to maximizing likelihood, since the logarithm is a monotonic function.

i.i.d. sample, the **log likelihood** is

$$l(\theta) = \sum_{i=1}^{n} \log[f(X_i|\theta)]$$

Maximum Likelihood

Ex: Let $x_1, x_2, \ldots, x_n$ be a sequence of n Bernoulli trials with the same p. $x_i \in \{0, 1\}, \forall i$

Each trial, the probability of observing x is $p^x(1-p)^{1-x}$

$$L(p) = \prod_{i=1}^{n} p^{x_i}(1-p)^{1-x_i}$$

$$= p^{\Sigma x_i}(1-p)^{n-\Sigma x_i}$$

$$L'(p) = (\Sigma x_i)p^{\Sigma x_i - 1}(1-p)^{n-\Sigma x_i} - (n - \Sigma x_i)p^{\Sigma x_i}(1-p)^{n-\Sigma x_i - 1}$$

$$p^{\Sigma x_i}(1-p)^{n-\Sigma x_i}\left(\frac{\Sigma x_i}{p} - \frac{n - \Sigma x_i}{1-p}\right) = 0$$

$$\frac{\Sigma x_i}{p} - \frac{n - \Sigma x_i}{1-p} = 0 \quad \Rightarrow \quad \hat{p} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

Maximum Likelihood

Ex: Bernoulli, working with log-likelihood

$$L(p) = \prod_{i=1}^{n} p^{x_i}(1-p)^{(1-x_i)}$$

$$\ell(p) = \log p \sum_{i=1}^{n} x_i + \log(1-p) \sum_{i=1}^{n} (1-x_i)$$

$$\frac{\partial \ell(p)}{\partial p} = \frac{\sum_{i=1}^{n} x_i}{p} - \frac{\sum_{i=1}^{n} (1-x_i)}{1-p}$$

$$\sum_{i=1}^{n} x_i - p \sum_{i=1}^{n} x_i = p \sum_{i=1}^{n} (1-x_i)$$

$$\widehat{p} = \frac{1}{n} \sum_{i=1}^{n} x_i$$

Maximum Likelihood

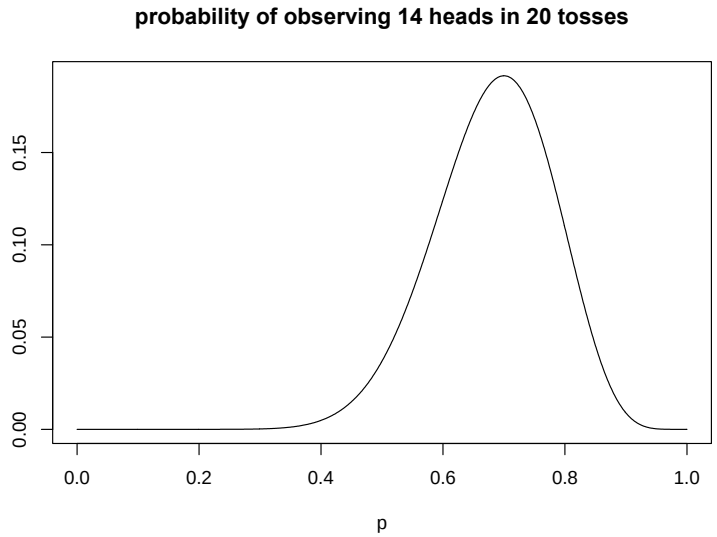
Ex: Binomial, n tosses, x heads

$$L(\theta) = P(x|\theta) = \binom{n}{x} p^x (1-p)^{n-x}$$

$$l(\theta) = log\binom{n}{x} + xlogp + (n-x)\log(1-p)$$

$$l'(\theta) = \frac{x}{p} - \frac{n-x}{1-p}$$

$$\hat{p} = \frac{x}{n}$$



Maximum Likelihood

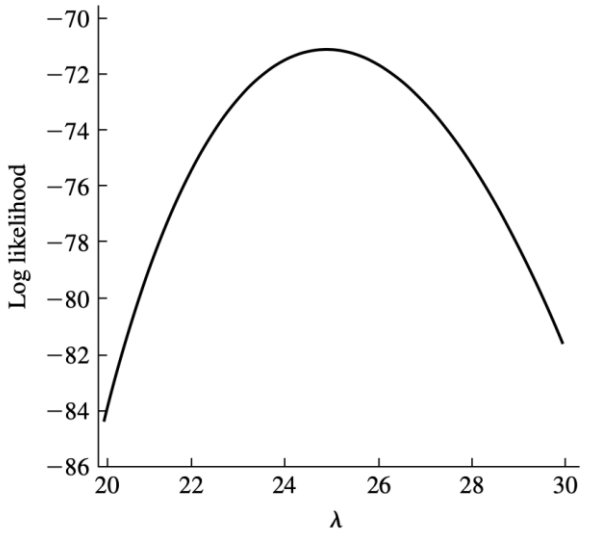
Ex: Poisson

If X follows a Poisson distribution with parameter λ, then

$$P(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}$$

If $X_1, \ldots, X_n$ are i.i.d. and Poisson,

$$\begin{aligned} l(\lambda) &= \sum_{i=1}^{n} (X_i \log \lambda - \lambda - \log X_i!) \\ &= \log \lambda \sum_{i=1}^{n} X_i - n\lambda - \sum_{i=1}^{n} \log X_i! \end{aligned}$$



Setting the first derivative of the log likelihood equal to zero,

$$l'(\lambda) = \frac{1}{\lambda} \sum_{i=1}^{n} X_i - n = 0 \quad \Rightarrow \quad \hat{\lambda} = \overline{X}$$

Maximum Likelihood

Ex: Normal

If $X_1, X_2, \ldots, X_n$ are i.i.d. $N(\mu, \sigma^2)$,

$$f(x_1, x_2, \ldots, x_n|\mu, \sigma) = \prod_{i=1}^{n} \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left[\frac{x_i - \mu}{\sigma}\right]^2\right)$$

$$l(\mu, \sigma) = -n\log\sigma - \frac{n}{2}\log 2\pi - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(X_i - \mu)^2$$

$$\frac{\partial l}{\partial \mu} = \frac{1}{\sigma^2}\sum_{i=1}^{n}(X_i - \mu) \qquad \frac{\partial l}{\partial \sigma} = -\frac{n}{\sigma} + \sigma^{-3}\sum_{i=1}^{n}(X_i - \mu)^2$$

$$\hat{\mu} = \overline{X} \longrightarrow \hat{\sigma} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X})^2}$$